

Jinzhou Tang

San Diego, CA | jit027@ucsd.edu | +1 (858) 220-6201

[Website](#) | [GitHub](#) | [Publications](#)

Research Interests

Embodied AI; interactive world models; self-improving agents; robot learning; visuomotor reasoning and learning from experience.

Education

University of California, San Diego

M.S. in Computer Science

Sep 2025 – Present

San Diego, CA

Research advisors: Hao Su and Biwei Huang.

Sun Yat-sen University

B.S. in Intelligent Science and Technology (GPA: 3.8/4.0)

Sep 2021 – Jul 2025

Shenzhen, China

Research Experience

Aether AI Technologies, Inc.

Research Intern – Embodied AI and World Models

Feb 2026 – Present

San Diego, CA

- Co-led **Video2World** (joint first author), an open benchmark for autonomous video-to-simulation with **222 reconstruction instances**, 39 task families, and 189 embodied demonstration videos.
- Developed and evaluated coding-agent workflows that construct executable simulators from robot and human videos; assessed geometric fidelity, dynamic fidelity, and functional success across **nine agent systems**.
- Worked on human-to-robot data conversion and action-conditioned world modeling for scalable robot learning and closed-loop model improvement.

HCP Lab, Sun Yat-sen University

Research Assistant – Embodied AI and Computer Vision

Jun 2023 – Jun 2025

Shenzhen, China

- Researched hierarchical embodied agents and computer vision; contributed to **HiVA** and foundation-model-based medical-imaging research, including **VertFound**.

Meituan

Research Intern – Embodied Reasoning and Human Behavior Modeling

Aug 2024 – Feb 2025

Beijing, China

- Developed methods for spatiotemporal reasoning and hierarchical, goal-conditioned behavior modeling; participated in method design, implementation, empirical evaluation, and paper preparation.
- Built hierarchical annotations and evaluation pipelines for long-horizon human motion and behavior generation.

CoRe Lab, Peking University

Research Intern – Video Understanding

May 2024 – Aug 2024

Beijing, China

- Worked with Yixin Zhu’s group on video understanding in biological laboratories, contributing to benchmark data, object detection/tracking, and baseline evaluations.

Selected Publications and Preprints

* Equal contribution. Peer-reviewed venue or preprint status is indicated for each entry.

- [1] **Video2World: Benchmarking Coding Agents for Interactive World Modeling from Embodied Videos**
Jinzhou Tang*, Zijun Zhang*, Jing Yang, Yuchen Yan, Kun Zhou, Lingjun Mao, Ruobing Han, Jinglin Cao, Wenpeng Xu, Lukun He, Minghao Fu, Fan Feng, Biwei Huang.
arXiv:2610.04432, 2026. [Paper] [Project] [Code] [Dataset]
- [2] **World Action Verifier: Self-Improving World Models via Forward-Inverse Asymmetry**
Yuejiang Liu*, Fan Feng*, Lingjing Kong*, Weifeng Lu*, Jinzhou Tang, Kun Zhang, Kevin Murphy, Chelsea Finn, Yilun Du.
NeurIPS 2026 (to appear). Outstanding Paper, ICLR 2026 Workshop on World Models. [Paper] [Project]
- [3] **DreamSAC: Learning Hamiltonian World Models via Symmetry Exploration**
Jinzhou Tang, Fan Feng, Minghao Fu, Wenjun Lin, Biwei Huang, Keze Wang.
CVPR 2026. [Paper] [Code]

- [4] **LASAR: Towards Spatio-temporal Reasoning with Latent Cognitive Map**
Jinzhou Tang, Sidi Liu, Waikit Xiu, Weixing Chen, Keze Wang.
CVPR 2026. [Paper] [Code]
- [5] **HiVA: Self-organized Hierarchical Variable Agent via Goal-driven Semantic-Topological Evolution**
Jinzhou Tang*, Jusheng Zhang*, Qinhan Lv*, Sidi Liu, Jing Yang, Chengpei Tang, Keze Wang.
AAAI 2026. [Paper] [Code]
- [6] **Mind2Dialogue: Training Human-Aware Language Models by Simulating User Mental States**
Zixuan Wang*, Yufan Zhou*, Jinzhou Tang*, Xinle Yu, Chengjun Wu, Lyumanshan Ye, Zhaoxiang Feng, Letian Peng, Adyasha Patra, Fan Bai, Enze Ma, Zhengding Hu, Jianyang Gu, Zhao Wang, Yufei Ding, Jingbo Shang, Tianmin Shu, Zhiting Hu, Zhen Wang.
arXiv:2609.15972, 2026. Accepted at NeurIPS 2026 UserSim and BiAlign Workshops. [Paper] [Project]
- [7] **OSC: Cognitive Orchestration through Dynamic Knowledge Alignment in Multi-Agent LLM Collaboration**
Jusheng Zhang, Yijia Fan, Kaitong Cai, Jinzhou Tang, Xiaofei Sun, Keze Wang.
Findings of EMNLP 2025. [Proceedings PDF]
- [8] **From Motion to Behavior: Hierarchical Modeling of Humanoid Generative Behavior Control**
Jusheng Zhang, Jinzhou Tang, Sidi Liu, Mingyan Li, Sheng Zhang, Jian Wang, Keze Wang.
arXiv:2506.00043, 2025. [Paper]
- [9] **VertFound: Synergizing Semantic and Spatial Understanding for Fine-grained Vertebrae Classification via Foundation Models**
Yinhao Wu*, Jinzhou Tang*, Zequan Yao, Mingjie Li, Yuan Hong, Dongdong Yu, Zhifan Gao, Bin Chen, Shen Zhao.
MICCAI 2024 (Early Accept). [Paper]

Selected Research Projects

Video2World – Agentic Video-to-Simulation Benchmark

2026

- Formulated a benchmark for automatically constructing executable interactive worlds from embodied videos; released public agent interfaces, datasets, evaluators, and reproducible code.
- Built a unified evaluation covering scene geometry, motion dynamics, and task-level functionality, enabling direct comparison across reconstruction agents.

EvoWAM-0 – Closed-Loop World Action Models for Minecraft

2026

- Released a reproducible closed-loop agent combining next-second video prediction and inverse dynamics to turn imagined observations into Minecraft controls, without hand-scripted skills. [Code]

Robot Shelf Organization with Franka Research 3

2025

- Integrated language-conditioned visual perception, grasp detection, and robot-arm motion planning using Molmo, AnyGrasp, MoveIt, and ROS 2 for shelf organization.

Technical Skills

Languages: Python, C/C++, Go, JavaScript, C#. **ML:** PyTorch, TensorFlow.

Robotics/Tools: ROS 2, MoveIt, Linux, Git, Docker; robot manipulation and simulation evaluation.

Selected Honors

2026: Outstanding Paper, ICLR Workshop on World Models (co-authored work).

2024: MICCAI Early Accept (VertFound); First-Class Award, National Statistical Modeling Contest (school level).

2023: Huawei Intelligent Foundation Scholarship; Second-Class Award, National Mathematical Modeling Contest (provincial).